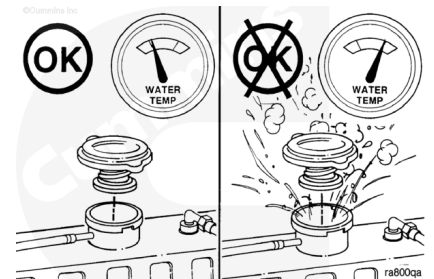


Trainings ▾

Drain

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



The QSK45 and QSK60 industrial engines use a two-pump/two-loop cooling system. The primary radiator provides cooling for the engine using the water pump mounted on the right-bank side of the engine. An LTA water pump is mounted on the left-bank side of the engine on the front gear cover. The LTA water pump circulates coolant through the two-pass core of the LTA radiator and back through the aftercoolers. The two center thermostats control temperature as the coolant passes through the thermostat housing.

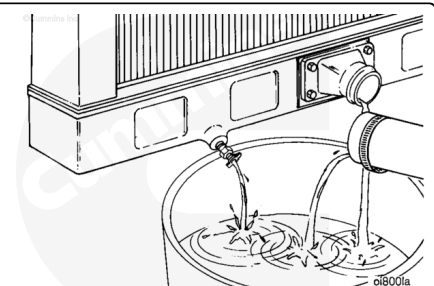
Remove the radiator cap after the engine is cool.

Open the petcocks on each turbocharger.

Aftercooled Engines

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



This cooling system uses the same radiator to cool the aftercoolers as is used to cool the engine. Refer to Procedure 200-003 in Section F.

(/qs3/pubsys2/xml/en/procedures/56/56-200-003.html)

Open the draincock at the bottom of the radiator.

Remove the lower radiator hose.

Drain the cooling system.

Do **not** allow the cooling system to dry out.

LTA

⚠ WARNING ⚠

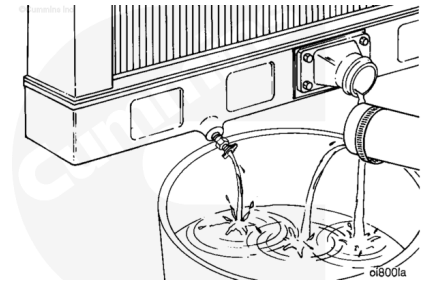
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

This system cools the aftercoolers, using a radiator separate from that used to cool the engine. Refer to Procedure 200-003 in Section F. (</qs3/pubsys2/xml/en/procedures/56/56-200-003.html>)

Drain the coolant.

The low-temperature aftercooling (LTA) cooling system is drained by opening the draincock in the LTA radiator or disconnecting the piping.

Do **not** allow the cooling system to dry out.

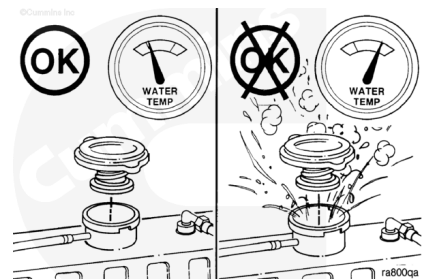


Flush

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

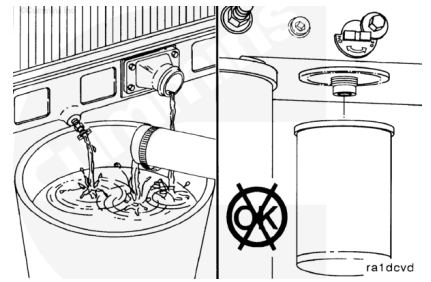
Drain the cooling system.



The performance of Restore™ depends on time, temperature, and concentration levels. An extremely scaled or flow-restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning time. Restore™ can be safely used up to twice the recommended concentration levels. Extremely scaled or fouled systems can require more than one cleaning.



Do **not** remove the coolant filters.



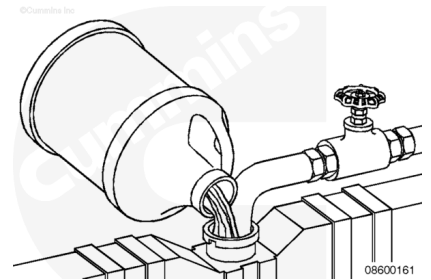
⚠ CAUTION ⚠

Fleetguard® Restore™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.

Close all petcocks to avoid air entrapment in the cooling system.

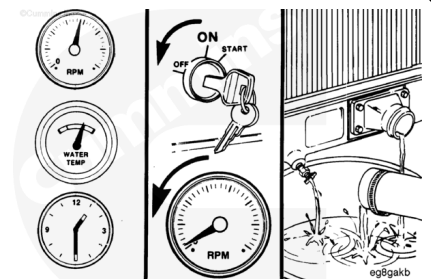
Immediately add 3.8 liters [1 gal] of Fleetguard® Restore™, or equivalent, for each 38 to 57 liters [10 to 15 gal] of cooling system capacity, and fill the system with plain water.

Turn the heater temperature switch to HIGH to allow maximum coolant flow through the heater core. The blower does **not** have to be on.

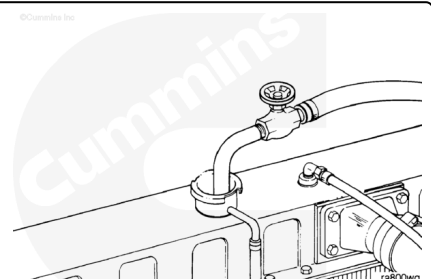


Operate the engine at a coolant temperature of at least 85°C [185°F] for 1 to 1-½ hours.

Shut the engine OFF, allow to cool to 50°C [120°F], and drain the cooling system.



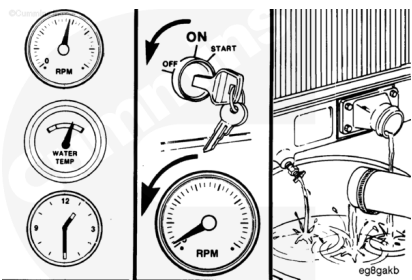
Fill the cooling system with clean water.



Operate the engine at high idle for 5 minutes with the coolant temperature above 85°C [185°F].

Shut the engine OFF, allow to cool to 50°C [120°F], and drain the cooling system.

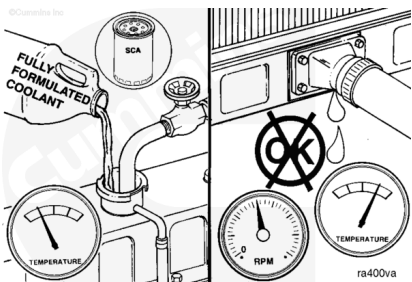
If the water drained is still dirty, the system **must** be flushed again until the water is clean.



Install a new coolant filter.

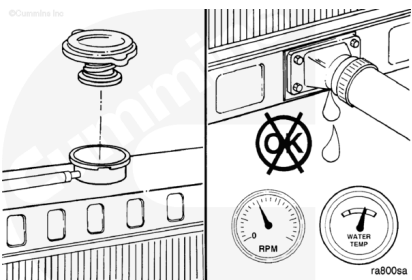
Fill the cooling system with fully formulated coolant.

Use additional SCA to bring the coolant to the correct SCA concentration level. Refer to Procedure 018-018 in Section V. (</qs3/pubsys2/xml/en/procedures/56/56-018-018.html>)



Install the pressure cap. Operate the engine until the coolant reaches a temperature of 70°C [158°F] and check for coolant leaks.

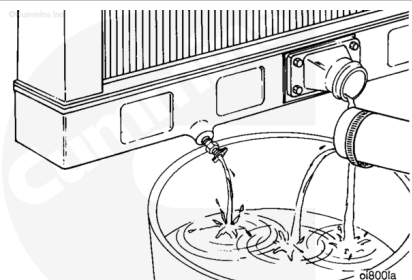
Measurements		
	kpa	psi
Minimum Cooling System Pressure Cap Capacity	76	11



Fill

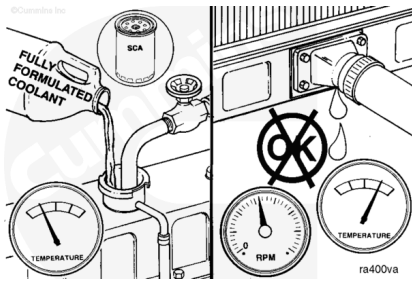
Make sure the radiator draincocks are closed.

Open the cooling system venting petcocks.



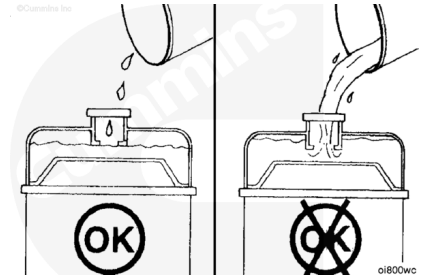
Use fully formulated coolant to fill the cooling system.

Use the correct units of SCA to obtain the correct cooling system protection. Refer to Section V.



⚠ CAUTION ⚠

Fill the cooling system slowly to avoid trapping air in the cooling system. Air trapped in the system can cause water pump cavitation and inadequate coolant level.



Begin filling the cooling system with coolant.

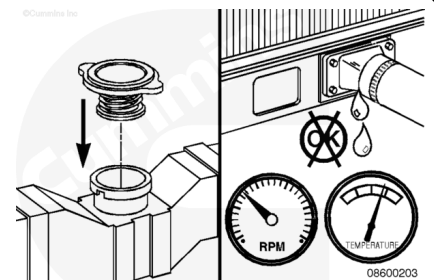
Fill until a continuous stream of coolant begins to flow from each opened petcock in the coolant system.

Close each petcock as the continuous stream of coolant begins to flow from that petcock.

Fill the system with the coolant to the bottom of the radiator fill neck.

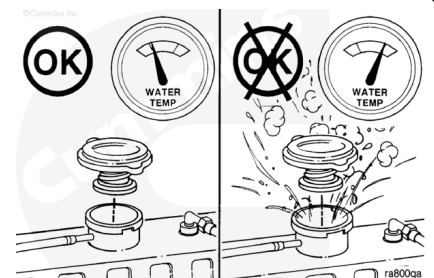
Install the cooling system pressure cap.

Operate the engine until the coolant reaches a temperature of 85°C [185°F]. Check for leaks.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Shut the engine off, and allow it to cool.

Check the coolant level.

